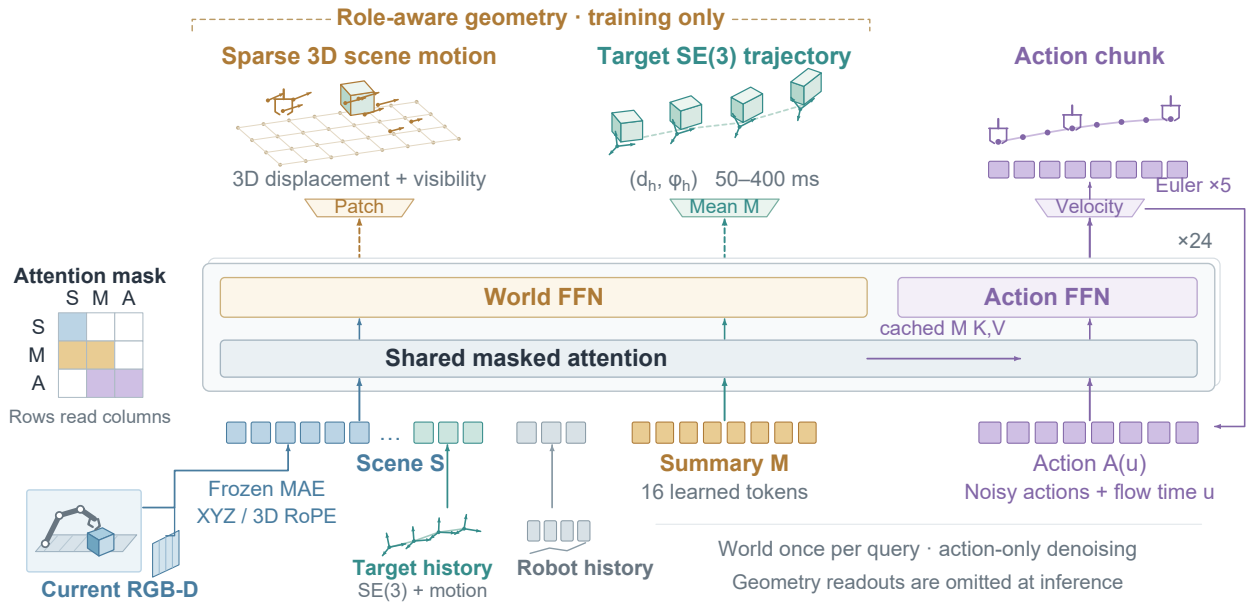

Role-aware geometry · training only
Sparse 3D scene motion
3D displacement + visibility
Patch
Target SE(3) trajectory
$(d_h, \varphi_h)$ 50–400 ms
Mean M
Action chunk
Euler ×5
Velocity
×24
Attention mask
S M A
S
M
A
Rows read columns
World FFN
Action FFN
cached M K,V
Shared masked attention
Scene S
Summary M
16 learned tokens
Action A(u)
Noisy actions + flow time u
Frozen MAE
XYZ / 3D RoPE
Current RGB-D
Target history
SE(3) + motion
Robot history
World once per query · action-only denoising
Geometry readouts are omitted at inference